

Lecture 8: Central Bank Balance Sheet and Money Supply Process

Instructor: Fei Tan

 @econdojo  @BusinessSchool101  Saint Louis University

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Overview

This lecture develops an understanding of how the central bank interacts with the financial system and how its balance sheet is connected to the money and credit that flows through the economy.

Key insights:

- Changes in the balance sheet affect the monetary base
- The money multiplier links the monetary base to the money supply
- Understanding these connections is essential for monetary policy analysis

The Road Ahead

1. Central Bank's Balance Sheet
2. Deposit Expansion Multiplier
3. Monetary Base and Money Supply

Central Bank's Balance Sheet

Central bank activities cause changes in its balance sheet:

- Supplying currency
- Providing deposit accounts to government and commercial banks
- Making loans
- Buying and selling securities and foreign currency

Figure 17.1 The Central Bank's Balance Sheet

	Assets	Liabilities
Government's Bank	Securities Foreign exchange reserves	Currency Government's account
Bankers' Bank	Loans	Accounts of the commercial banks (reserves)

Assets of Central Bank

1. Securities

- Primary assets of most central banks
- Controlled through open market operations (purchases and sales)

2. Foreign exchange reserves

- Central bank's and government's balances of foreign currency
- Held as bonds issued by foreign governments
- Used for foreign exchange interventions

3. Loans

- To commercial banks and nonbanks
- Service to commercial banks

Liabilities of Central Bank

1. Currency

- Principal liability of most central banks
- Circulating in the hands of the nonbank public

2. Government's account

- Government deposits funds (primarily tax revenues)
- Government writes checks and makes electronic payments

3. Commercial bank accounts (reserves)

- Sum of deposits at central bank plus vault cash
- Required reserves: Mandated by regulation
- Excess reserves: Held voluntarily

Note: Monetary Base (high-powered money) = Currency + Reserves

Open Market Operations

The Fed buys or sells securities in financial markets.

Example: Fed purchases \$1 billion T-bills from a commercial bank

- Fed's assets and liabilities both increase by \$1 billion → monetary base ↑ by \$1 billion
- Banking system: no change in liabilities, changes in assets sum to zero

Figure 17.2

Balance Sheet Changes after the Federal Reserve Purchases a U.S. Treasury Bond

A. Federal Reserve's Balance Sheet		B. Banking System's Balance Sheet	
Assets	Liabilities	Assets	Liabilities
Securities (U.S. Treasury bond) +\$1 billion	Reserves +\$1 billion	Reserves +\$1 billion	
		Securities (U.S. Treasury bond) -\$1 billion	

Foreign Exchange Intervention

The Fed buys or sells foreign currency reserves.

Example: Fed purchases \$1 billion worth of euros (buying German government bonds)

- Fed's assets and liabilities both increase by \$1 billion → monetary base ↑ by \$1 billion
- Banking system: no change in liabilities, changes in assets sum to zero

Figure 17.3

Balance Sheet Changes after the Federal Reserve Purchases a German Government Bond

A. Federal Reserve's Balance Sheet		B. Banking System's Balance Sheet	
Assets	Liabilities	Assets	Liabilities
Foreign exchange reserves +\$1 billion (German government bonds in euros)	Reserves +\$1 billion	Reserves +\$1 billion Securities −\$1 billion (German government bonds)	

Discount Loans

The Fed makes loans backed by collateral (assets pledged by borrower).

Example: Fed makes \$100 million loan to a commercial bank

- Fed's assets and liabilities both increase by \$100 million → monetary base ↑ by \$100 million
- Banking system: assets and liabilities both increase by \$100 million → balance sheet expands

Figure 17.4 Balance Sheet Changes after the Federal Reserve Makes a Discount Loan

A. Federal Reserve's Balance Sheet				B. Banking System's Balance Sheet			
Assets		Liabilities		Assets		Liabilities	
Discount loans	+\$100 million	Reserves	+\$100 million	Reserves	+\$100 million	Discount loans	+\$100 million

Cash Withdrawal

Cash withdrawal by the nonbank public affects only the composition.

Example: Withdraw \$100 from an ATM

- Fed: currency outstanding $\uparrow$ \$100, reserves $\downarrow$ \$100 $\rightarrow$ composition changes, monetary base size unchanged
- Banking system: reserves $\downarrow$ \$100, checkable deposits $\downarrow$ \$100 $\rightarrow$ balance sheet shrinks

Figure 17.5

Balance Sheet Changes after a Private Person Withdraws Cash from His or Her Bank Account

A. Nonbank Public's Balance Sheet			
Assets		Liabilities	
Currency	+\$100		
Checkable deposits	-\$100		

B. Federal Reserve's Balance Sheet		C. Banking System's Balance Sheet	
Assets		Assets	
		Reserves	-\$100
Liabilities		Liabilities	
Currency	+\$100	Checkable deposits	-\$100
Reserves	-\$100		

Deposit Expansion Process

Consider a \$100,000 open market purchase by the Fed from First Bank.

Simplifying assumptions:

- Banks hold no excess reserves
- No change in currency held by nonbank public

Process:

1. First Bank receives \$100,000 in reserves
2. First Bank keeps required reserves, lends out the rest
3. Borrowed funds deposited in Second Bank
4. Second Bank keeps required reserves, lends out the rest
5. Process continues...

First Bank Balance Sheet

Figure 17.6

Changes in First Bank's Balance Sheet after the Fed's Purchase of a U.S. Treasury Bond

A. Immediate Impact			
Assets		Liabilities	
Reserves	+\$100,000		
Securities	-\$100,000		

B. After the Extension of a Loan		C. After Withdrawal by the Borrower	
Assets		Assets	
Reserves	+\$100,000	Reserves	\$0
Securities	-\$100,000	Securities	-\$100,000
Loans	+\$100,000	Loans	+\$100,000

		Liabilities	
		Checkable deposits	+\$100,000

		Liabilities	
		Checkable deposits	\$0

Initial impact:

- Reserves increase by \$100,000
- Securities decrease by \$100,000

After lending:

- Reserves kept = required reserves
- Excess reserves lent out

Second Bank Balance Sheet

Figure 17.7

Changes in Balance Sheets

A. Second Bank after American Steel's Deposit				B. Second Bank after Extension of a Loan			
Assets		Liabilities		Assets		Liabilities	
Reserves	+\$100,000	American Steel's checking account	+\$100,000	Reserves	+\$10,000	American Steel's checking account	+\$100,000
				Loan	+\$90,000		

C. Third Bank after Deposit and Extension of a Loan			
Assets		Liabilities	
Reserves	+\$ 9,000	Checking account	+\$90,000
Loan	+\$81,000		

When borrower from First Bank deposits funds:

- Second Bank receives deposits
- Second Bank makes new loans after keeping required reserves
- Process continues through banking system

Deposit Multiplier

Let r_D be the **required reserve ratio**. A one-dollar increase in reserves creates a deposit increase of:

$$1 + (1 - r_D) + (1 - r_D)^2 + (1 - r_D)^3 + \dots = \frac{1}{r_D} > 1$$

Table 17.3

Multiple Deposit Expansion following a \$100,000 Open Market Purchase
Assuming a 10% Reserve Requirement

Bank	Increase in Deposits	Increase in Loans	Increase in Reserves
First Bank	\$ 0	\$ 100,000	\$ 0
Second Bank	\$ 100,000	\$ 90,000	\$ 10,000
Third Bank	\$ 90,000	\$ 81,000	\$ 9,000
Fourth Bank	\$ 81,000	\$ 72,900	\$ 8,100
Fifth Bank	\$ 72,900	\$ 65,610	\$ 7,290
Sixth Bank	\$ 65,610	\$ 59,049	\$ 6,561
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The Banking System	\$1,000,000	\$1,000,000	\$100,000

Monetary Base and Money Supply

Now we relax the assumptions:

- Banks may hold excess reserves
- Currency held by public may change

We derive the **money multiplier** m : the relationship between the quantity of money and the monetary base.

$$M = m \times MB$$

where

- $M = C + D$ (money = currency + deposits)
- $MB = C + R$ (monetary base = currency + reserves)
- $R = RR + ER$ (reserves = required + excess)

Money Multiplier

$$M = \underbrace{\frac{C/D + 1}{C/D + r_D + ER/D}}_{\text{money multiplier } m} \times MB$$

Define two important ratios:

1. Excess reserve-to-deposit ratio: ER/D

- Cost: Interest on loans forgone
- Benefit: Safety if deposits withdrawn suddenly

2. Currency-to-deposit ratio: C/D

- Cost: Interest on deposits forgone
- Benefit: Lower risk and greater liquidity

These ratios are determined by economic agents' optimization decisions.